Lösung: Grenzwerte mit Regel von l'Hospital

a) $\lim\limits_{x\to 0} \frac{\sin(x)}{x} \quad \left(= \frac{"0"}{"0"}\right)$

$$\lim_{x\to 0} \frac{\sin(x)}{x} = \lim_{x\to 0} \frac{(\sin(x))'}{(x)'} = \lim_{x\to 0} \frac{\cos(x)}{1}$$

$$= \lim_{x\to 0} \cos(x) = \cos(0) = \underline{\underline{1}}$$

b) $\lim\limits_{x\to 0} \frac{1 - 1/\cos(x)}{x^2} \quad \left(= \frac{"0"}{"0"}\right)$

$$\lim_{x\to 0} \frac{1 - 1/\cos(x)}{x^2} = \lim_{x\to 0} \frac{(1 - 1/\cos(x))'}{(x^2)'}$$

$$= \lim_{x\to 0} \frac{-\frac{1}{\cos^2(x)} \cdot (-\sin(x))}{2x} = \lim_{x\to 0} \frac{\sin(x)}{2x \cdot \cos^2(x)} \quad \left(= \frac{"0"}{"0"}\right)$$

$$= \lim_{x\to 0} \frac{(\sin(x))'}{(2x \cdot \cos^2(x))'} = \lim_{x\to 0} \frac{\cos(x)}{2\cdot\cos^2(x) + 2x \cdot 2 \cdot \cos(x) \cdot (-\sin(x))}$$

$$= \lim_{x\to 0} \frac{\cos(x)}{2\cdot\cos(x)(\cos(x) - 2\cdot x\sin(x))}$$

$$= \frac{\cos(0)}{2\cdot\cos(0)(\cos(0) - 2\cdot 0\sin(0))} = \underline{\underline{1/2}}$$

c) $\lim\limits_{x\to 0} \frac{\ln(2x+1)}{x} \quad \left(= \frac{"0"}{"0"}\right)$

$$\lim_{x\to 0} \frac{\ln(2x+1)}{x} = \lim_{x\to 0} \frac{(\ln(2x+1))'}{(x)'} = \lim_{x\to 0} \frac{\frac{2}{2x+1}}{1}$$

$$= \lim_{x\to 0} \frac{2}{2x+1} = \frac{2}{1} = \underline{\underline{2}}$$

d) $\lim\limits_{x\to 0} \dfrac{\arctan(x)}{x}$ $\left(= \dfrac{\text{“}0\text{”}}{\text{“}0\text{”}}\right)$

$$\lim_{x\to 0} \frac{\arctan(x)}{x} = \lim_{x\to 0} \frac{(\arctan(x))'}{(x)'} = \lim_{x\to 0} \frac{\frac{1}{1+x^2}}{1}$$

$$= \lim_{x\to 0} \frac{1}{1+x^2} = \frac{1}{1+0^2} = \underline{\underline{1}}$$

e) $\lim\limits_{x\to \infty} \dfrac{x^3}{e^x}$ $\left(= \dfrac{\text{“}\infty\text{”}}{\text{“}\infty\text{”}}\right)$

$$\lim_{x\to \infty} \frac{(x^3)'}{(e^x)'} = \lim_{x\to \infty} \frac{3x^2}{e^x} \quad \left(= \frac{\text{“}\infty\text{”}}{\text{“}\infty\text{”}}\right)$$

$$= \lim_{x\to \infty} \frac{(3x^2)'}{(e^x)'} = \lim_{x\to \infty} \frac{6x}{e^x} \quad \left(= \frac{\text{“}\infty\text{”}}{\text{“}\infty\text{”}}\right)$$

$$= \lim_{x\to \infty} \frac{(6x)'}{(e^x)'} = \lim_{x\to \infty} \frac{6}{e^x} = \underline{\underline{0}}$$

Die e-Funktion setzt sich (immer) gegen den polynomiellen Term durch.